Mutation Testing Report v1 (Draft)

COMP 441 Software Analysis and Testing - Group Project "AI-Augmented Test Suite" - Track C (Healthcare)

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1. Summary

The AI-only baseline suite (118 tests) was mutation-tested with cosmic-ray on two modules. Across **27 mutants, 17 were killed (63.0%)** and **10 survived**. patient/services.py scored 100%, while patient/forms/patients.py scored only **37.5%**. Of the 10 survivors, 2 reveal real behavioural gaps (an untested default argument and an untested defensive branch) and 8 are presentation-only (textarea display attributes). Manual tests that kill all 10 have been drafted and are pending verification. This is a v1 snapshot: patient/models.py and the remaining modules are not yet mutated.

2. Setup and method

Item	Detail
Mutation tool	cosmic-ray 8.7.0 (mutmut was not used: WSL was unavailable on the test machine)
Environment	Windows, Python 3.12.10, Django 4.2.9, pytest 9.1.1, pytest-cov 7.1.0, pytest-django 4.14.0
Suite under test	`tests/ai_generated/test_patient_ai.py` only (AI-generated, prompt #1, unedited). 118 tests: 117 passed, 1 xfailed.
Test command per mutant	`pytest -x -q tests/ai_generated -k <module-specific filter> -p no:cacheprovider` - narrowed to the tests that exercise the mutated module so each mutant runs in seconds.
Modules mutated so far	`patient/services.py` (filter: extract_keywords); `patient/forms/patients.py` (filter: registration_form or search_form)
Mutation score	killed mutants / total mutants (a timeout or incompetent mutant would be counted separately; none occurred)

3. AI baseline context

Statement coverage was 11% before AI (commit fc1e5b9) and 21% after adding the AI-only suite (run of 25 Oct 2026: 6,021 statements, 4,759 missed, measured across all project packages). This figure includes migrations and management commands, which are normally untested, so a fairer adjusted figure will be reported in the final report. One AI test is marked xfail and records a genuine defect: the generated patient ID (format P-YYYY-NNNN, 11 characters) exceeds the 10-character `patient_id` limit. SQLite does not enforce the limit, so a stricter database would reject or truncate the value.

4. Results

Module	Mutants	Killed	Survived	Mutation score
`patient/services.py`	11	11	0	100.0%
`patient/forms/patients.py`	16	6	10	37.5%
Combined	**27**	**17**	**10**	**63.0%**

patient/services.py (all killed): mutants on `extract_common_keywords` included comparison-operator replacements on the word-length boundary (`len(word) > 2`), an `and` to `or` replacement in the stopword filter, numeric-constant replacements and a zero-iteration loop. The AI suite has explicit boundary tests for word length, default limit and limit of zero, which is why every mutant was caught.

Reporting note: the raw forms score is 37.5%. If the 8 presentation-only survivors are excluded under a stated rule (no effect on behaviour), the adjusted forms score is 6 of 8, or 75%. The raw figure is the headline; the adjusted figure is secondary.

5. Surviving mutants: root-cause analysis (patient/forms/patients.py)

#	Mutation	Why the AI-only suite missed it	Test-design technique	Manual test (drafted)
1	`save(self, commit=True)` changed to `commit=False` (ReplaceTrueWithFalse 0)	Every AI save test passes *commit* explicitly, so the default argument is never exercised.	Equivalence partitioning: argument omitted vs supplied	`test_save_without_argument_persists_patient`: calls `save()` and asserts the patient is persisted
2	`and` changed to `or` in the `medical_conditions` / `hasattr` check (ReplaceAndWithOr 0)	AI tests only exercise the case where both conditions are true. The defensive branch (field absent from cleaned data) is never reached.	Decision table / condition coverage, case (False, True)	`test_save_skips_mapping_when_medical_conditions_missing`: removes the field, then asserts `save(commit=False)` does not raise (mutant raises KeyError)
3-10	Textarea `rows` values changed on four fields (NumberReplacer 0-7), detailed below	AI tests assert what the form accepts and rejects. None assert display attributes. Behaviour is unchanged, so these are presentation-only, not equivalent, mutants.	UI-contract check (parametrised)	`test_textarea_widgets_have_expected_rows`: 4 parameters kill all 8 mutants

Detail of mutants 3-10 (original value 2 or 3 changed by one):

Mutant	Field	Original	Mutated
NumberReplacer 0	medical_conditions	rows 2	rows 3
NumberReplacer 1	medical_conditions	rows 2	rows 1
NumberReplacer 2	address	rows 3	rows 4
NumberReplacer 3	address	rows 3	rows 2
NumberReplacer 4	allergies	rows 2	rows 3
NumberReplacer 5	allergies	rows 2	rows 1
NumberReplacer 6	chronic_diseases	rows 2	rows 3
NumberReplacer 7	chronic_diseases	rows 2	rows 1

6. Limitations and honest caveats

Only 27 mutants across 2 modules have been run, so the 63.0% figure is not representative of the whole system. The test command is narrowed per module, so a mutant is only tested against the relevant AI tests, not the full suite. The manual tests in section 5 are drafted but have not yet been run against the mutants; the v2 score will be recorded only after they pass and the mutation run is repeated. Eight of the ten survivors are cosmetic, so more substantive survivors are expected from patient/models.py (age calculation, patient ID generation).

7. Next steps

Step	Status
Verify the three manual tests pass, then re-run mutation testing on patient/forms/patients.py (v2 data point)	Pending
Mutate patient/models.py (age calculation, patient ID, full name) and annotate survivors; target at least 8 meaningful case studies in total	Pending
AI-guided / risk-based test prioritisation order with justification (required Week 7 item)	Not started

Step	Status
Re-run on the final hybrid suite for v1 to v2 chart (Week 11)	Planned (3 Nov)

Appendix A: evidence files (repository, branch maano_branch)

File	Content
`ai-baseline/ai_only_run_2026-10-25.txt`	AI-only suite run with coverage across all project packages
`ai-baseline/mutation_v1_services.txt`	cosmic-ray report, patient/services.py (11 jobs, 0 survivors)
`ai-baseline/mutation_v1_forms.txt`	cosmic-ray report, patient/forms/patients.py (16 jobs)
`ai-baseline/mutation_v1_forms_survivors.txt`	Survivors with code diffs
`cosmic-ray-services.toml, cosmic-ray-forms.toml`	Mutation run configurations (reproducible)

Reproduce: `cosmic-ray init cosmic-ray-forms.toml forms.sqlite; cosmic-ray baseline cosmic-ray-forms.toml; cosmic-ray exec cosmic-ray-forms.toml forms.sqlite; cr-rate forms.sqlite`

AI tool use: this draft was prepared with assistance from Claude (analysis structure and drafting). The group must review and verify all findings and re-express the root-cause explanations in its own words before final submission, and disclose this in the Declaration of AI Tool Use.